Autoencoders in Machine Learning

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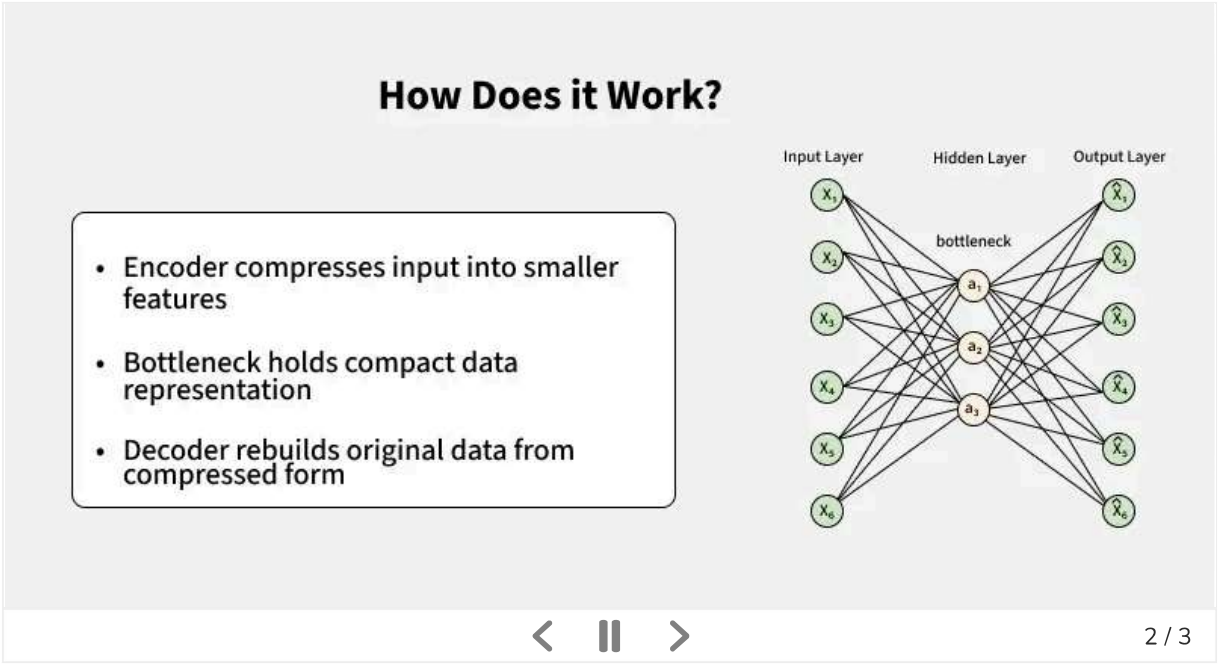
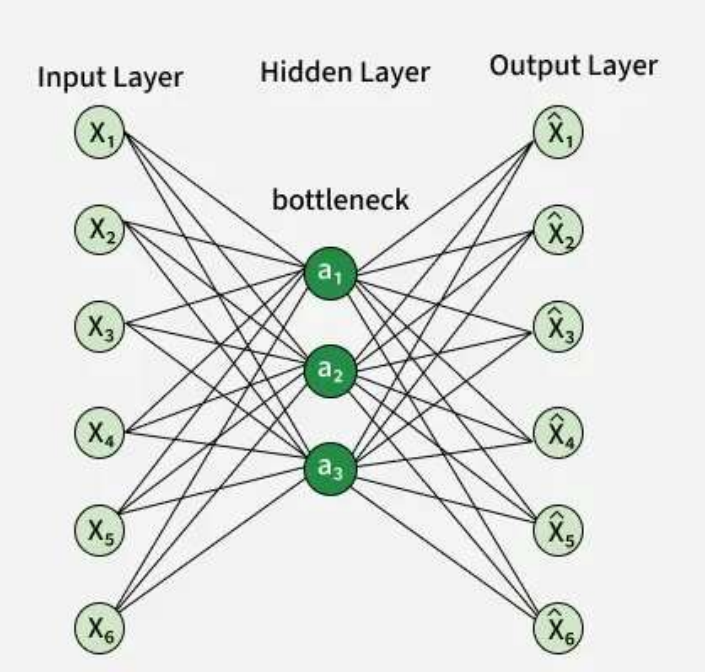
Autoencoders are neural networks that compress input data into a smaller representation and then reconstruct it, helping the model learn important patterns efficiently. Their key uses include:

- Helps remove unwanted noise from data and improve quality
- Identifies unusual patterns or anomalies in data
- Extracts important features for better model performance



Architecture of Autoencoder

An autoencoder's architecture consists of three main components that work together to compress and then reconstruct data which are as follows:



Autoencoder

1. Encoder

It compress the input data into a smaller, more manageable form by reducing its dimensionality while preserving important information. It has three layers which are:

- **Input Layer**: Here the original data enters the network. It can be images, text features or any other structured data.
- **Hidden Layers**: These layers perform a series of transformations on the input data. Each hidden layer applies weights and activation functions to capture important patterns, progressively reducing the data's size and complexity.
- **Output (Latent Space)**: The encoder outputs a compressed vector known as the latent representation or encoding. This vector captures the important features of the input data in a condensed form helps in filtering out noise and redundancies.

2. Bottleneck (Latent Space)

The bottleneck is the smallest layer in the network that holds a compressed representation of the input data. It forces the model to keep only the most important features, helping it learn key patterns and improve generalization.

3. Decoder

It is responsible for taking the compressed representation from the latent space and reconstructing it back into the original data form.

- **Hidden Layers**: Progressively expand the latent vector back into a higher-dimensional space. Through successive transformations decoder attempts to restore the original data shape and details
- **Output Layer**: Produces the reconstructed output which aims to closely resemble the original input. The quality of reconstruction depends on how well the encoder-decoder pair can minimize the difference between the input and output during training.

Loss Function in Autoencoder Training

During training an autoencoder's goal is to minimize the reconstruction loss which measures how different the reconstructed output is from the original input. The choice of loss function depends on the type of data being processed:

- **Mean Squared Error (MSE)**: Commonly used for continuous data. It measures the average squared differences between the input and the reconstructed data.
- **Binary Cross-Entropy**: Used for binary data (0 or 1 values). It calculates the difference in probability between the original and reconstructed output.

Efficient Representations in Autoencoders

Autoencoders learn compact and meaningful representations by applying constraints during training. After training, the encoder can be used to generate efficient feature representations for similar data.

- **Small Hidden Layers:** Forces the network to focus on important features and reduce redundancy
- **Regularization:** Uses L1 or L2 penalties to prevent overfitting and improve generalization
- **Denoising:** Adds noise during training so the model learns robust, noise-free features
- **Activation Function Tuning:** Promotes sparsity by activating only relevant neurons, reducing complexity

Types of Autoencoders

Lets see different types of Autoencoders which are designed for specific tasks with unique features:

1. Denoising Autoencoder

Denoising Autoencoder is trained to handle corrupted or noisy inputs, it learns to remove noise and helps in reconstructing clean data. It prevent the network from simply memorizing the input and encourages learning the core features.

2. Sparse Autoencoder

Sparse Autoencoder contains more hidden units than input features but only allows a few neurons to be active simultaneously. This sparsity is controlled by zeroing some hidden units, adjusting activation functions or adding a sparsity penalty to the loss function.

3. Variational Autoencoder

Variational autoencoder (VAE) makes assumptions about the probability distribution of the data and tries to learn a better approximation of it. It uses stochastic gradient descent to optimize and learn the distribution of latent variables. They used for generating new data such as creating realistic images or text.

It assumes that the data is generated by a Directed Graphical Model and it learns an approximate posterior $q_\phi(z|x)$ and a likelihood $p_\theta(x|z)$ where ϕ and θ are the parameters of the encoder and decoder respectively.

4. Convolutional Autoencoder

Convolutional autoencoder uses convolutional neural networks (CNNs) which are designed for processing images. The encoder extracts features using convolutional layers and the decoder reconstructs the image through deconvolution also called as upsampling.

Implementation

We will create a simple autoencoder with two Dense layers:

- Encoder that compresses images into a 64-dimensional latent vector.
- Decoder that reconstructs the original image from this compressed form.

Step 1: Import necessary libraries

We will be using Matplotlib, NumPy, TensorFlow and the MNIST dataset loader for this.

```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers, losses
from tensorflow.keras.models import Model
from keras.datasets import mnist
```

Step 2: Load the MNIST dataset

We will be loading the MNIST dataset which is inbuilt dataset and normalize pixel values to [0,1] also reshape the data to fit the model.

```
(x_train, _), (x_test, _) = mnist.load_data()
```

```
(x_train, _), (x_test, _) = mnist.load_data()

x_train = x_train.astype('float32') / 255.
x_test = x_test.astype('float32') / 255.

x_train = np.reshape(x_train, (len(x_train), 28, 28, 1))
x_test = np.reshape(x_test, (len(x_test), 28, 28, 1))
```

Output:

Shape of the training data: (60000, 28, 28)
Shape of the testing data: (10000, 28, 28)

Step 3: Define a basic Autoencoder

Creating a simple autoencoder class with an encoder and decoder using Keras Sequential model.

- **layers.Input(shape=(28, 28, 1))**: Input layer expecting grayscale images of size 28x28.
- **layers.Dense(latent_dimensions, activation='relu'):** Dense layer that compresses the input to the latent space using ReLU activation.
- **layers.Dense(28 * 28, activation='sigmoid'):** Dense layer that expands the latent vector back to the original image size with sigmoid activation.

```
class SimpleAutoencoder(Model):
    def __init__(self, latent_dimensions):
        super(SimpleAutoencoder, self).__init__()
        self.encoder = tf.keras.Sequential([
            layers.Input(shape=(28, 28, 1)),
            layers.Flatten(),
            layers.Dense(latent_dimensions, activation='relu'),
        ])

        self.decoder = tf.keras.Sequential([
            layers.Dense(28 * 28, activation='sigmoid'),
            layers.Reshape((28, 28, 1))
        ])

    def call(self, input_data):
        encoded = self.encoder(input_data)
        decoded = self.decoder(encoded)
        return decoded
```

Step 4: Compiling and Fitting Autoencoder

Here we compile the model using Adam optimizer and Mean Squared Error loss also we train for 10 epochs with batch size 256.

- **latent_dimensions = 64**: Sets the size of the compressed latent space to 64.

```
latent_dimensions = 64
autoencoder = SimpleAutoencoder(latent_dimensions)
autoencoder.compile(optimizer='adam', loss=losses.MeanSquaredError())
```

```
autoencoder.fit(x_train, x_train,
                epochs=10,
                batch_size=256,
                shuffle=True,
                validation_data=(x_test, x_test))
```

Output:

```
Epoch 1/10
235/235 ━━━━━━━━━━━━━━━━━━━━ 3s 10ms/step - loss: 0.0973 - val_loss: 0.0321
Epoch 2/10
235/235 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 0.0289 - val_loss: 0.0209
Epoch 3/10
235/235 ━━━━━━━━━━━━━━━━━━━━ 2s 8ms/step - loss: 0.0197 - val_loss: 0.0151
Epoch 4/10
235/235 ━━━━━━━━━━━━━━━━━━━━ 3s 10ms/step - loss: 0.0144 - val_loss: 0.0115
Epoch 5/10
235/235 ━━━━━━━━━━━━━━━━━━━━ 3s 10ms/step - loss: 0.0112 - val_loss: 0.0093
Epoch 6/10
235/235 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 0.0091 - val_loss: 0.0077
Epoch 7/10
235/235 ━━━━━━━━━━━━━━━━━━━━ 2s 8ms/step - loss: 0.0076 - val_loss: 0.0067
Epoch 8/10
235/235 ━━━━━━━━━━━━━━━━━━━━ 2s 8ms/step - loss: 0.0066 - val_loss: 0.0059
Epoch 9/10
235/235 ━━━━━━━━━━━━━━━━━━━━ 3s 9ms/step - loss: 0.0060 - val_loss: 0.0055
Epoch 10/10
235/235 ━━━━━━━━━━━━━━━━━━━━ 3s 10ms/step - loss: 0.0056 - val_loss: 0.0051
```

Training

Step 5: Visualize original and reconstructed data

Now compare original images and their reconstructions from the autoencoder.

- **encoded_imgs = autoencoder.encoder(x_test).numpy()**: Passes test images through the encoder to get their compressed latent representations as NumPy arrays.
- **decoded_imgs = autoencoder.decoder(encoded_imgs).numpy()**: Reconstructs images by passing the latent representations through the decoder and converts them to NumPy arrays.

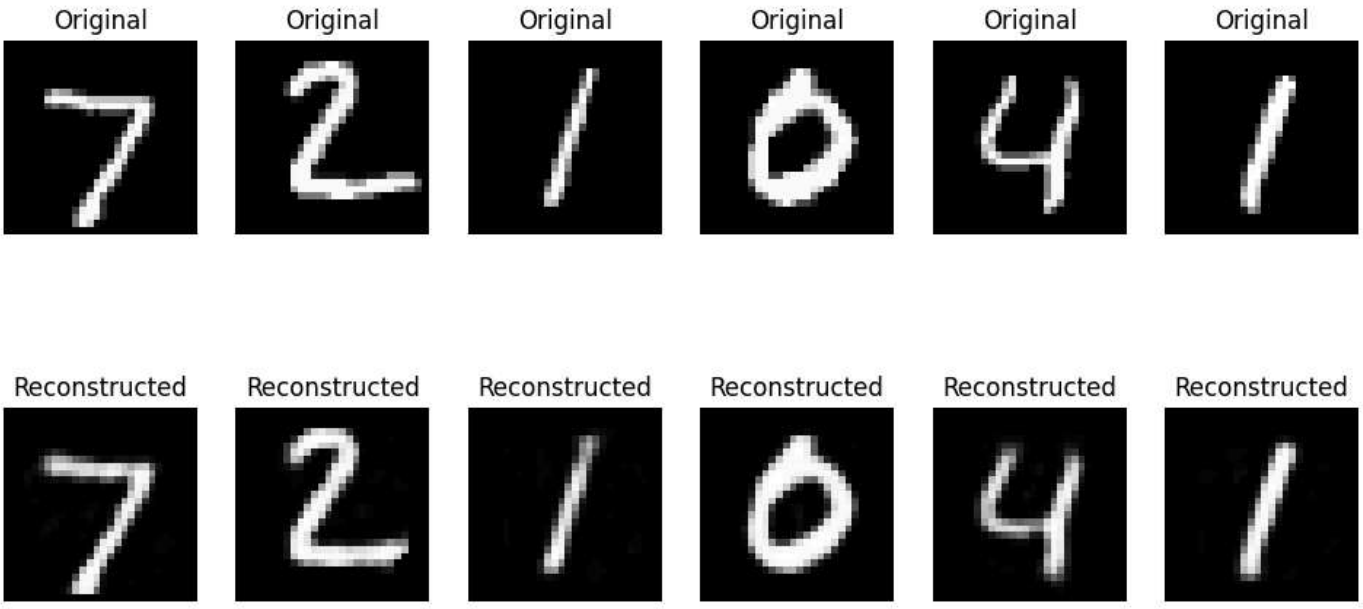
```
encoded_imgs = autoencoder.encoder(x_test).numpy()
decoded_imgs = autoencoder.decoder(encoded_imgs).numpy()

n = 6
plt.figure(figsize=(12, 6))
for i in range(n):
    ax = plt.subplot(2, n, i + 1)
    plt.imshow(x_test[i].reshape(28, 28), cmap='gray')
    plt.title("Original")
    plt.axis('off')

    ax = plt.subplot(2, n, i + 1 + n)
    plt.imshow(decoded_imgs[i].reshape(28, 28), cmap='gray')
    plt.title("Reconstructed")
    plt.axis('off')
```

```
plt.show()
```

Output:



The visualization compares original MNIST images (top row) with their reconstructed versions (bottom row) showing that the autoencoder effectively captures key features despite some minor blurriness.

Limitations

- Sometimes memorize the training data rather than learning meaningful patterns which reduces their ability to generalize to new data.
- Output may be blurry or distorted with noisy inputs or if the model architecture lacks sufficient complexity to capture all details.
- Require large amounts of data and careful parameter tuning (latent dimension size, learning rate, etc) to perform well. Insufficient data or poor tuning can result in weak feature representations.

Suggested Quiz 5 Questions

What is the primary goal of an autoencoder?

A Reduce data dimensionality while preserving information

B Generate new samples from learned patterns

C Assign input data to predefined categories

D Reconstruct input data from a compressed encoding

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